# The Fundamental Theorem of Arithmetic.

Every natural number greater than 1 factors uniquely into primes.

Ex. $14779666 = 2 \cdot 11 \cdot 11 \cdot 157 \cdot 389$
$= 2 \cdot 11^2 \cdot 157 \cdot 389.$

Proof of Existence. We proceed via strong induction. The base case is 2. 2 is prime, so $2=2$ is our desired factorization. Now suppose we have $n \geq 2$ & prime factorizations for all $2 \leq x \leq n$. We wish to show $n+1$ has a prime factorization. If $n+1$ is not divisible by any $x$ with $2 \leq x \leq n$, then $1$ & $n+1$ are the only positive divisors of $n+1$. Hence $n+1$ is prime, and $n+1 = n+1$ is our desired prime factorization.

Conversely, suppose $n$ is divisible by some $d$ with $2 \leq d \leq n$.

Thus $n+1 = jk$ where $2 \le j, k \le n$. By the induction hypothesis $j$ & $k$ have prime factorizations:

$$j = \prod_{i=1}^{r} p_i^{e_i} \quad \& \quad k = \prod_{j=1}^{s} q_j^{f_j}$$

$p_1^{e_1} \cdot p_2^{e_2} \cdots p_r^{e_r}$

where the $p_i$ $1 \le i \le r$ are distinct primes

as are the $q_j$ $1 \le j \le s$.

We see $n+1 = jk = \prod_{i=1}^{r} p_i^{e_i} \cdot \prod_{j=1}^{s} q_j^{f_j}$.

Thus we have achieved our desired prime factorization.

Strong induction completes the proof.

Ex.

$$j = 14779666 = 2 \cdot 11 \cdot 11 \cdot 157 \cdot 389$$
$$= 2 \cdot 11^2 \cdot 157 \cdot 389$$
$$= p_1^1 \, p_2^2 \, p_3^1 \, p_4^1$$

$e_2 = 2$ & $e_1 = 1$

$k = 24 = 2^3 \cdot 3$

$q_1 = 2 \quad q_2 = 3$

$f_1 = 3 \quad f_2 = 1$

$n+1 = 14779666 \cdot 24$

$$= \prod_{i=1}^{4} p_i^{e_i} \cdot \prod_{j=1}^{2} q_i^{f_i}$$

$$= 2 \cdot 11^2 \cdot 157 \cdot 389 \cdot 2^3 \cdot 3$$

$$2 \cdot 11 \cdot 11 \cdot 157 \cdot 389 \cdot 2 \cdot 2 \cdot 2 \cdot 3$$

$\sum$

Sum

$\prod$

product

# Proof of Uniqueness.

BWOC suppose prime factorization isn't unique.

By well-ordering we can pick $n$, the smallest positive integer without a prime factorization.

Suppose $n = \prod_{i=1}^{r} p_i = \prod_{j=1}^{s} q_j$ are two distinct prime factorizations of $n$.

Let $p = p_1$. We see $p \mid \prod_{j=1}^{s} q_j$.

That is, $p \mid q_1 \cdot (q_2 \cdots q_s)$.

By Euclid, $p \mid q_1$ or $p \mid (q_2 \cdots q_s)$.

If $p \mid q_1$, then $p = q_1$.

Thus $n = p \prod_{i=2}^{r} p_i = p \cdot \prod_{j=2}^{s} q_j$

We see $\frac{n}{p} = \prod_{i=2}^{r} p_i = \prod_{j=2}^{s} q_j$.

This contradicts $n$ being the smallest w/ 2 prime factorizations.

Big picture idea of proof.

We have $n = p \cdot p_2 \cdots p_r = q_1 \cdots q_s$. Euclid tells us $p \mid q_1$ or $p \mid (q_2 \cdots q_s)$.

If $p \mid q_1$, then $p = q_1$ and we end up contradicting the minimality of $n$.

If $p \mid (q_2 \cdots q_s)$, then $p \mid q_2$ or $p \mid (q_3 \cdots q_s)$.

If $p \mid q_2$, then $p = q_2$ and we end up contradicting the minimality of $n$.

If $p \mid (q_3 \cdots q_s)$, then $p \mid q_3$ or $p \mid (q_4 \cdots q_s)$.

$\vdots$

We either finish or we end up with $p \mid q_{s-1} q_s$.

Euclid says $p \mid q_{s-1}$ or $p \mid q_s$.

In either case we have a contradiction.